

Hybrid Search Architecture

Goal

Combine semantic, keyword, and graph search.

Architecture

```
graph TD; A[User Query] --> B[Query Rewriter]; B --> C[Vector Search]; B --> D[Graph Search]; B --> E[BM25 Search]; C --> F[Reranker]; D --> F; E --> F; F --> G[Context Fusion]; G --> H[LLM]; H --> I[Response];
```

User Query
↓
Query Rewriter
↓
Vector Search
Graph Search
BM25 Search
↓
Reranker
↓
Context Fusion
↓
LLM
↓
Response

Vector Search

Database

Qdrant

Embedding

BGE-Large

Purpose

Semantic similarity

Weight

40%

Graph Search

Database

Neo4j

Purpose

Relationship traversal

Weight

40%

BM25 Search

Purpose

Keyword lookup

Weight

20%

Final Ranking Formula

```
Final Score =  
0.4 × Vector Score +  
0.4 × Graph Score +  
0.2 × BM25 Score
```

Query Workflow

Example

Why did Pump P101 fail?

Execution

```
Query
↓
Entity Detection
↓
Qdrant Search
↓
Neo4j Traversal
↓
BM25 Search
↓
Context Fusion
↓
Reranker
↓
Gemini
↓
Answer
```

Context Fusion

Merge

- Graph Context
- Vector Context
- Keyword Context
- Memory Context

Reranker

Model

BAAI BGE-Reranker-v2

Purpose

Improve Top-K quality

Failure Handling

Qdrant Down

↓

Neo4j + BM25

Neo4j Down

↓

Vector Search

All Failed

↓

Fallback LLM